

NIRAV P SHETTY

B.Tech CSE, MIT Manipal · Batch 2027 · CGPA: 6.2 (62%)
shettynirav2005@gmail.com · Phone: 9380236186 ; 7996975557
[Portfolio](#) · [GitHub](#) · [LinkedIn](#)



PROFESSIONAL SUMMARY

Computer Science undergraduate (Batch 2027, MIT Manipal) with 2 internships and 4+ deployed full-stack and AI-native projects. Independently built a RAG-grounded, self-healing AI test automation platform (FastAPI, local Llama 3 via Ollama, Playwright) later deployed in production during an internship, plus a Supabase/PostgreSQL app with row-level security and a 3-service enterprise ERP platform with Nginx gateway and CI/CD. Builds daily with AI coding assistants (GitHub Copilot, Claude, Cursor) and is drawn to agentic systems and AI-native engineering.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, HTML5, CSS3, C++, C

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Socket.io, Tailwind CSS, Bootstrap

Backend & Databases: PostgreSQL (Supabase), MongoDB, REST APIs, Scalable API Design

AI & LLM: LLM Integration (Ollama/Llama 3, Gemini API), RAG-Style Grounding, Prompt Engineering, Semantic Search (Sentence-Transformer Embeddings), AI Coding Assistants (GitHub Copilot, Claude, Cursor, Codex)

Cloud & Tools: Nginx, Git, GitHub, Vercel, AWS (Forage simulation)

WORK EXPERIENCE

IGS Global — Software Testing Intern

Completed

- **Deployed a self-built AI test automation platform** (FastAPI, local Llama 3 via Ollama, Playwright — see QA Copilot Dashboard) into live testing workflows, generating and self-healing test suites directly from plain-English user stories.
- **Auto-repaired broken Playwright locators** via sentence-transformer embeddings and live DOM matching and built a log-clustering pipeline segmenting user sessions into behavior personas.

GoPerch Innovations Pvt. Ltd. — Full stack developer intern

Completed

- **Delivered 7+ ERP inventory management features** into a live production system, streamlining workflows and improving usability across multiple business modules.
- **Collaborated in a 10-developer agile team** — sprint planning, code reviews, and CI/CD deployment cycles — shipping scalable, production-ready code.

PROJECTS

QA Copilot Dashboard — Self-Healing Test Automation Platform · Python · FastAPI · React · Playwright · Ollama (Llama 3) · Locust

- **Architected a full-stack AI QA platform** (FastAPI + React + Playwright) that converts plain-English user stories into structured test cases (tabular + Gherkin/BDD), then generates runnable **Playwright scripts** grounded in the target site's live scraped HTML/DOM rather than LLM-guessed selectors - executed end-to-end with live-streamed (SSE) results.
- **Built a self-healing locator engine** using sentence-transformer embeddings and cosine similarity over a live DOM scrape to automatically detect and repair broken selectors at runtime, without manual test maintenance.
- **Designed a self-learning retrieval layer** that persists confirmed-correct selectors and retrieves them by embedding similarity to ground future test generation, improving accuracy over repeated runs with no fine-tuning or vector database.
- **Extended the platform with Jira ticket import, image-to-story OCR, and Locust-based load-test generation** (SLA thresholds, HTML reporting) — running entirely on a local Llama 3 model via Ollama, with zero cloud API calls.

Enterprise Management Software for Windscares [\[GitHub\]](#) · TypeScript · React · Python · Nginx · REST APIs · CI/CD

- **Built a 3-service ERP platform** (React/TypeScript, Python REST API, Nginx) with 5+ modules — inventory, finance, reporting, auth, dashboards.
- **Added RBAC and SSL via Nginx**, securing 100% of ingress through one gateway with 3 permission tiers.

NutriTrust — Smart Pantry & Expiry-Tracking App [\[Live\]](#) [\[GitHub\]](#) · Next.js · React · Supabase (PostgreSQL) · Gemini API

- **Built a household pantry-tracking app** (Next.js + Supabase/PostgreSQL) with shared shopping lists and scan history, using the Gemini API for OCR to parse receipts, labels, and expiry dates.
- **Designed the relational schema and wrote Row-Level Security policies and security-definer PL/pgSQL functions** for atomic multi-user operations (household creation, invite-code joins); debugged and resolved a production RLS infinite-recursion bug (Postgres 42P17).

Wrdl.io — Real-Time Multiplayer Wordle [\[Live\]](#) [\[GitHub\]](#) · Next.js · TypeScript · Socket.io

- **Built a Sub-second game-state sync across concurrent players** via Socket.io event-driven architecture — <1s latency, 0 client-side library dependencies.

CERTIFICATIONS

- Google AI Online Professional Certificate — Coursera [\[Certificate\]](#)
- Deloitte Australia Technology Job Simulation — Forage (Jul 2025) [\[Certificate\]](#)
- Designing Industry-Ready Scalable APIs & Database Systems — MSSP 2025 [\[Certificate\]](#)
- AWS APAC Solutions Architecture Virtual Experience — Forage (Jul 2025) [\[Certificate\]](#)